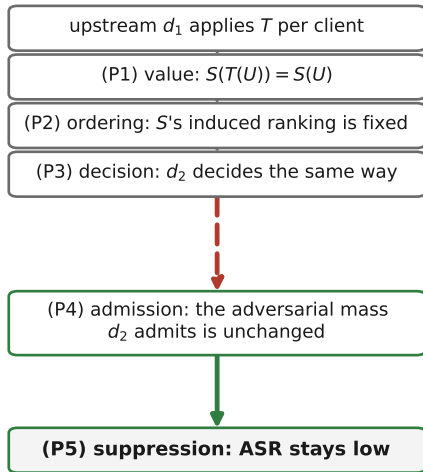




*(P1)  $\Rightarrow$  (P2)  $\Rightarrow$  (P3): implications, true by definition*

IT BREAKS HERE, IN BOTH DIRECTIONS

(P3) held, (P5) failed: cos\_krum's selection is bit-identical at every rung (decision, admission 0.000), yet ASR falls 0.425

(P3) failed, (P5) held: Krum's decision flips 0.80 of rounds with admission  $\equiv$  0.000 (adversarial share pinned at 0.267), and ASR stays flat (0.016–0.062)

**Statistic preservation is not security preservation: only the (P4)  $\rightarrow$  (P5) link carries security content in our experiments.**